

UNCLASSIFIED // FOR OFFICIAL USE ONLY (SYNTHETIC DEMO)

TECHNICAL MANUAL
OPERATION AND MAINTENANCE INSTRUCTIONS

EMERGENCY DIESEL GENERATOR (EDG) SET
Model EDG-450kW Shipboard Auxiliary Power Unit

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PREPARING ACTIVITY
VISION ONE TECH, LLC — SYNTHETIC DEMO MANUAL
Capability Package Demo Artifact — Performance Support Video Factory

DISTRIBUTION / CLASSIFICATION

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NOTICE: This publication is a **synthetic demonstration manual** created for AI pipeline evaluation. It is not an official U.S. Navy, DoD, or OEM technical publication. Equipment designations, procedures, and part numbers are fictional but styled after common shipboard technical manual (TM / NSTM) conventions.

1. REVISION HISTORY

Highlighted Rev B rows identify sections that trigger change-impact detection and selective re-render of dependent Performance Support videos.

Rev	Date	Description of Change	Affected Sections
A	15 Sep 2026	Initial release — baseline procedures for EDG-450kW set.	All
B (PLANNED)	01 Nov 2026	CHANGED: PROC-EDG-002 Step 6 torque from 35 ft-lb to 42 ft-lb ; mandatory post-change oil sample. CHANGED: PROC-EDG-003 coolant from 50/50 to 60/40 for arctic deployment. PROC-EDG-001 unchanged.	5.2, 5.3, Fig 5-3, Fig 5-5

NOTE: Authoritative content of this PDF is **Revision A**.

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3. EQUIPMENT DESCRIPTION

3.1 System Overview

The Emergency Diesel Generator (EDG) Model EDG-450kW provides automatic and manual emergency electrical power for critical shipboard loads when ship service generators are unavailable. The set consists of a turbocharged inline-six diesel engine coupled to a brushless synchronous alternator, mounted on a common skid with integral radiator cooling, starting battery bank, and local control panel (LCP).

Under automatic mode, the EDG starts on loss of preferred bus voltage, comes up to rated speed and voltage within 10 seconds, and closes the generator breaker onto the emergency bus.

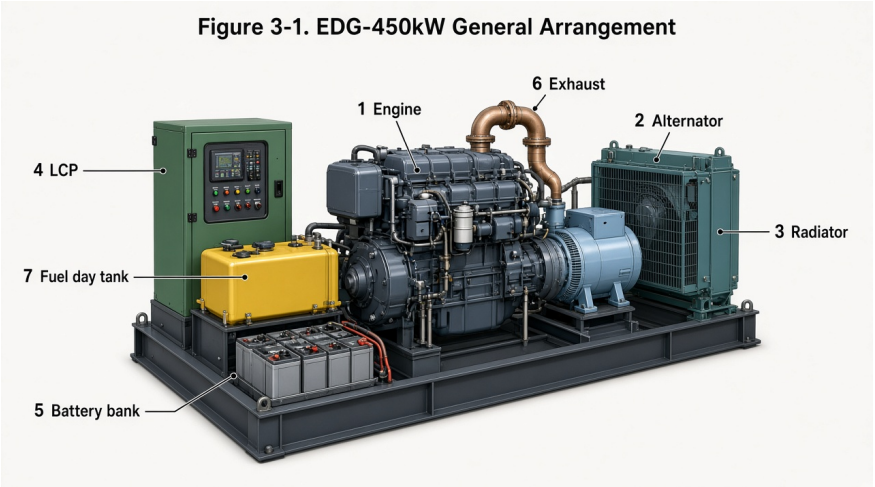


Figure 3-1. EDG-450kW set — general arrangement (engine, alternator, radiator, LCP, battery bank, exhaust, fuel day tank).

3.2 Major Assemblies

- **Diesel Engine:** Four-stroke, turbocharged, aftercooled, 1800 RPM continuous, electronic governor, wet-sump lubrication.
- **Alternator:** 450 kW, 0.8 PF, 450 VAC, 60 Hz, 3-phase, Class H insulation, permanent-magnet exciter with AVR.
- **Cooling System:** Engine-driven pump, thermostatic bypass, skid-mounted radiator with electric fan, expansion tank.
- **Fuel System:** Duplex primary filters, engine-mounted secondary filter, electric priming pump, day-tank float switches.
- **Local Control Panel:** Engine gauges, AC metering, start/stop, mode select, alarm annunciator, emergency stop.

3.3 Technical Characteristics

Parameter	Value
Rated Power	450 kW / 562.5 kVA
Voltage / Frequency	450 VAC, 60 Hz, 3 ϕ
Engine Speed	1800 RPM
Fuel	F-76 / DF-2 diesel
Lube Oil Capacity	28 L (7.4 gal) SAE 15W-40
Coolant Capacity	42 L (11.1 gal)
Starting System	24 VDC battery / starter
Dry Weight (approx.)	3,200 kg (7,055 lb)

4. SAFETY AND PRECAUTIONS

WARNING — PERSONNEL HAZARD

Rotating machinery can cause severe injury or death. Ensure EDG is tagged out (LOTO) per shipboard electrical safety procedures before performing maintenance. Verify zero energy: open generator breaker, isolate fuel and coolant valves as required.

WARNING — HOT SURFACES / FLUIDS

Engine, exhaust, and coolant may exceed 90 °C (194 °F) after operation. Allow cool-down or use approved PPE before opening filters, drain plugs, or radiator caps.

CAUTION — ENVIRONMENTAL

Contain and dispose of used oil, filters, and coolant per applicable hazardous material instructions. Do not discharge overboard.

All maintainers shall hold current qualifications for diesel machinery and electrical systems as required by the command training plan.

5. MAINTENANCE PROCEDURES

5.1 PROC-EDG-001 — Pre-Operational Inspection

Procedure ID	PROC-EDG-001
Title	Pre-Operational Inspection of EDG Set
Periodicity	Before each automatic-mode enable; weekly if standby
Estimated Time	15 minutes
Personnel	1 qualified electrician / engineman
Tools / Materials	Flashlight, absorbent pads, IR thermometer (optional), Form EDG-CHK-01
Figures	Figure 3-1, Figure 5-1

WARNING

Do not disable protective trips during inspection. If abnormal oil or coolant leakage is found, do not place the set in automatic mode.

Procedure Steps

1. Verify EDG mode switch on Local Control Panel is in OFF or MANUAL. Confirm generator breaker is OPEN.
2. Inspect skid base and drip pan for oil or coolant. Wipe minor seepage; report active leaks on Form EDG-CHK-01.
3. Check engine oil level on dipstick; oil shall be between ADD and FULL marks. Top off with SAE 15W-40 if required.
4. Visually inspect coolant level in expansion tank sight glass; level shall be between LOW and FULL.
5. Inspect primary and secondary fuel filter bowls for water. Drain water if present into an approved container.
6. Verify battery charger ON and battery voltage between 24.0 and 28.0 VDC at LCP meter.
7. Inspect air intake for debris; ensure rain hood and crankcase breather are clear.
8. Walk around exhaust flex and silencer; look for soot streaks indicating exhaust leaks.
9. Record results on Form EDG-CHK-01 and return mode switch to AUTOMATIC only if all checks are satisfactory.

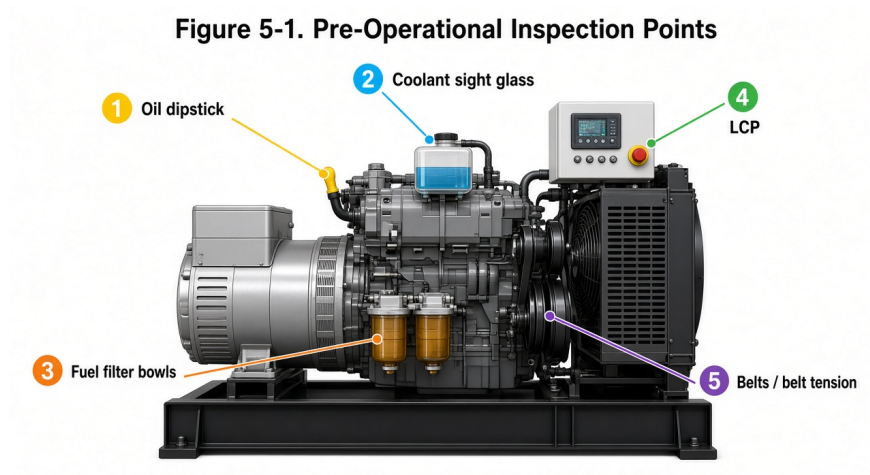


Figure 5-1. Pre-operational inspection points — oil dipstick, coolant sight glass, fuel filter bowls, LCP, belts / belt tension.

5.2 PROC-EDG-002 — Lube Oil Filter Replacement

Procedure ID	PROC-EDG-002
Title	Engine Lube Oil Filter Element Replacement
Periodicity	Every 250 operating hours or annually, whichever first
Estimated Time	45 minutes
Personnel	1 engineman; helper optional
Tools / Materials	Filter wrench, drain pan (20 L), torque wrench, filter P/N EDG-LOF-450, SAE 15W-40, gloves
Figures	Figure 5-2, Figure 5-3

WARNING — HOT OIL

Perform only after engine cool-down of at least 30 minutes, or wear approved thermal gloves.

Procedure Steps

1. Tag out EDG per LOTO: mode OFF, breaker OPEN, fuel rack locked as required.
2. Place drain pan under the spin-on lube oil filter. Position absorbent pads around the filter base (Figure 5-2).
3. Using filter wrench, loosen filter counterclockwise. Allow oil to drain, then remove filter.
4. Inspect old filter sealing face for debris. Discard filter in approved oily-waste container.
5. Clean filter mounting adapter face. Lightly lubricate the new filter gasket with clean engine oil.
6. Install new filter P/N EDG-LOF-450 hand-tight until gasket contacts adapter, then +3/4 turn. **Final torque: 35 ft-lb (Rev A).** Verify with torque wrench (Figure 5-3).
7. Check crankcase oil level; add SAE 15W-40 as needed to FULL mark.
8. Remove LOTO. Start engine in MANUAL unloaded for 2 minutes. Observe filter area for leaks.
9. Stop engine, recheck oil level after 5 minutes. Return set to AUTOMATIC if no leaks.
10. Annotate maintenance log with hours, filter PN, oil added, and technician ID.

Change-impact note: Rev B revises Step 6 torque to 42 ft-lb and adds mandatory oil sample.

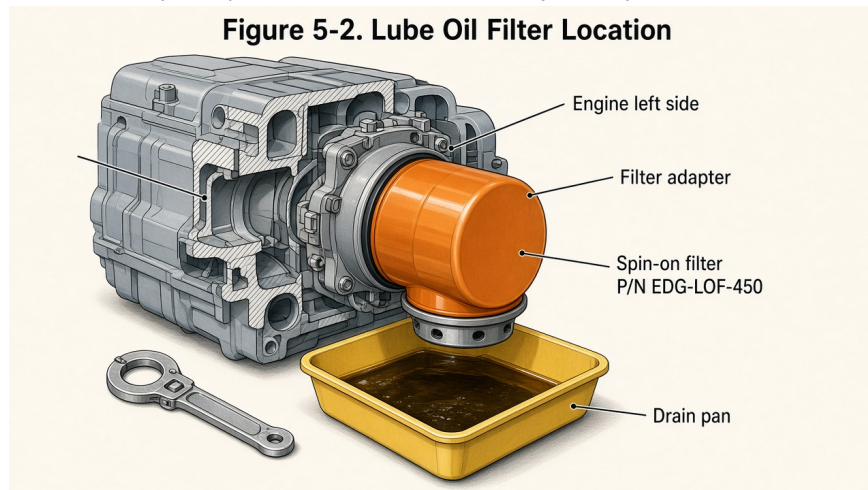


Figure 5-2. Lube oil filter location on engine left side, looking aft.

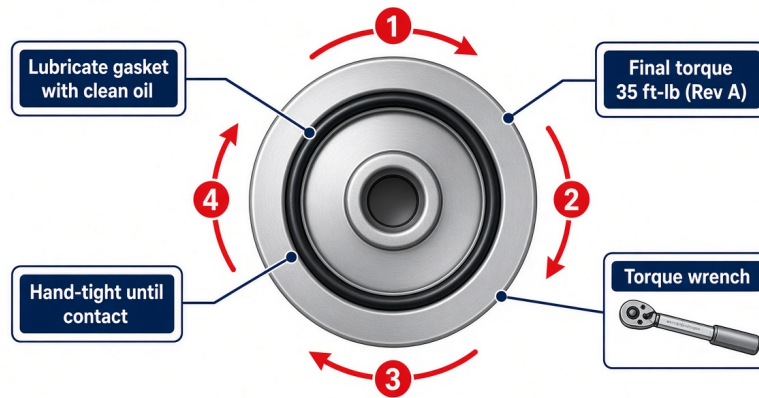
Figure 5-3. Gasket Lubrication and Torque Sequence

Figure 5-3. Filter installation — gasket lubrication and torque application points (35 ft-lb Rev A).

5.3 PROC-EDG-003 — Cooling System Service

Procedure ID	PROC-EDG-003
Title	Cooling System Drain, Flush, and Refill
Periodicity	Every 2 years or 2,000 hours; sooner if contamination found
Estimated Time	2.5 hours
Personnel	2 technicians recommended
Tools / Materials	Drain hose, catch containers (≥50 L), distilled water, propylene glycol, refractometer, PPE
Figures	Figure 5-4, Figure 5-5

WARNING — PRESSURIZED HOT COOLANT

Never remove the radiator cap while hot or pressurized. Allow full cool-down; open slowly with eye protection.

Procedure Steps

1. Tag out EDG. Confirm engine is cool (block temperature < 40 °C / 104 °F).
2. Place containers under radiator petcock and engine block drain (Figure 5-4).
3. Slowly open radiator cap, then open petcock and block drain. Allow complete drain.
4. Close drains. Fill with distilled water. Start MANUAL, circulate 10 minutes to flush.
5. Stop engine, drain flush water. Repeat once if fluid remains discolored.
6. Mix propylene glycol and distilled water to a **50/50** ratio (Rev A). Verify ≈ -37 °C (-34 °F) with refractometer (Figure 5-5).
7. Fill slowly through radiator neck. Bleed air at high-point bleeder if installed.
8. Install radiator cap. Run unloaded to thermostat open; recheck expansion tank after cool-down.
9. Inspect hoses and clamps for seepage before returning to AUTOMATIC.
10. Record coolant mixture, volume, freeze point, and technician ID in the maintenance log.

Change-impact note: Rev B changes Step 6 mixture to 60/40 for arctic deployment.

Figure 5-4. Coolant Drain Locations and Flow

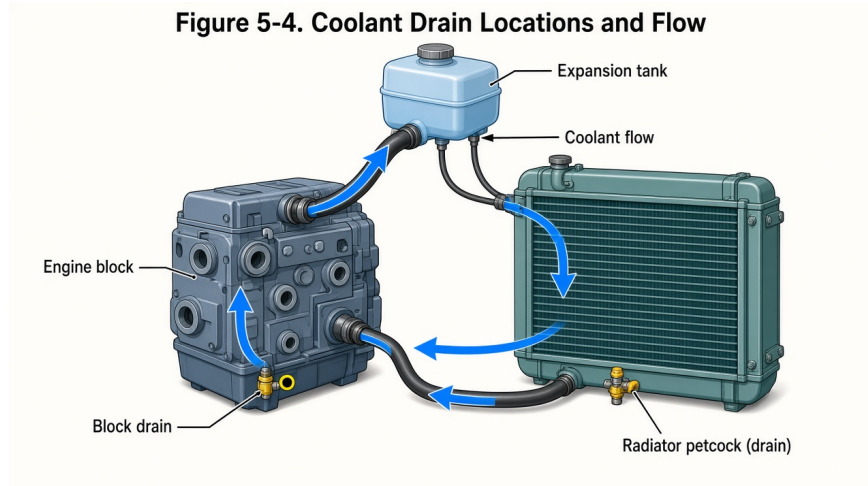


Figure 5-4. Radiator petcock and engine block drain locations.

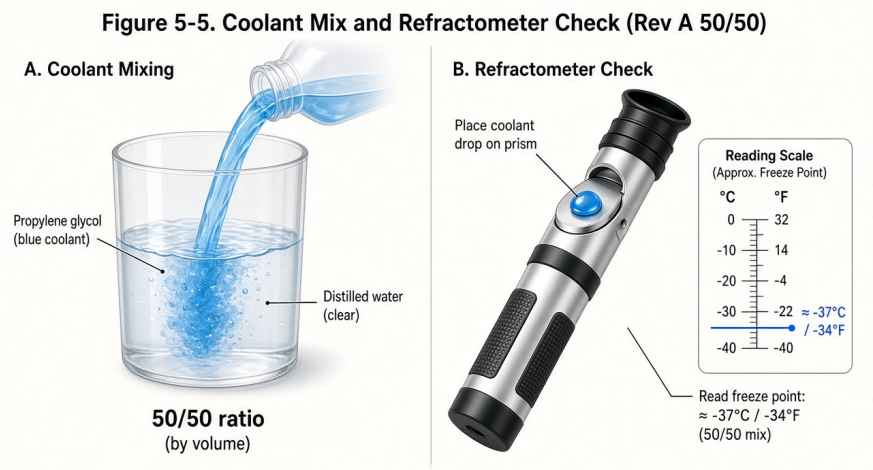


Figure 5-5. Coolant mixing and refractometer check — 50/50 propylene glycol (Rev A).

6. ILLUSTRATED PARTS / FIGURE LIST

Figure	Title
Fig 3-1	EDG-450kW general arrangement (incl. fuel day tank)
Fig 5-1	Pre-operational inspection points (dipstick, coolant, filters, LCP, belts)
Fig 5-2	Lube oil filter location (color cutaway)
Fig 5-3	Filter gasket lubrication and torque sequence 1–4
Fig 5-4	Coolant drain points with blue flow arrows
Fig 5-5	Coolant mix / refractometer check (two panels)

7. REFERENCES AND RELATED DOCUMENTS

- NAVSEA S9086-HN-STM-010 — NSTM Chapter 233, Diesel Engines (reference style only; not reproduced).
- OEM Publication EDG-450-OM — Operation Manual (synthetic companion).
- Form EDG-CHK-01 — Pre-Operational Inspection Checklist.
- Shipboard Hazardous Material Control Plan — used oil and coolant disposal.
- Vision One Performance Support Video Factory — demo lineage mapping for PROC-EDG-001/002/003.

END OF DOCUMENT — S9550-AB-MMA-010 Rev A — UNCLASSIFIED // FOR OFFICIAL USE ONLY (SYNTHETIC DEMO)

Synthetic demonstration artifact for AI Performance Support Video Factory evaluation.